ZCAS UNIVERSITY

SCHOOL OF COMPUTING AND APPLIED SCIENCES

BACHELOR OF INFORMATION TECHNOLOGY/BACHELOR OF NETWORK ENGINEERING

CCS3632 NETWORK MANAGEMENT AND SUPPORT

END OF SEMESTER FINAL EXAMINATION

FRIDAY 29th NOVEMBER 2024

09:00 – 12:00 HOURS

TIME ALLOWED: WRITING – THREE HOURS: READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This section is compulsory and must be attended.
2. Section B: Answer **THREE (3)** questions from this section.
3. This examination carries a total of **100 marks.**
4. Candidates must not turn this page until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

QUESTION 1.

a) A Liquid Telecom Network Engineer has network 140.8.0.0 to subdivide into other networks. He requires 5 subnets of equal size with approximately over 8,000 hosts each.

REQUIRED:

Provide the following information in the given order:

i. The number of network bits to obtain for the required number of subnets. **(2 Marks)**

ii. The number of hosts available per subnet after subnetting. **(4 Marks)**

iii. The decimal value of the new subnet mask. **(4 Marks)**

iv. Write the network in Slash Notation. **(4 Marks)**

v. The incremental value of the subnets. **(2 Marks)**

vi. The first host, last host, and broadcast of each subnet. **(9 Marks)**

b) You happen to work as a network support engineer at MBZ Ltd and you have been experiencing problems with your local area network of late. The CEO of MBZ Ltd has asked you to urgently resolve the problem as clients have been complaining about the poor service delivery.

You consulted your ISP, LIQUID Telecom which gave you a network: 195.100.50.0 /27 to help resolve the problem by using the subnetting method.

Required:

Using a table, provide the ranges for the first five subnets. Show your calculations. **(15 Marks)**

[Total: 40 Marks]

SECTION B: Attempt any THREE questions in this section.

QUESTION 2.

You have recently been hired as a network administrator for a medium-sized company that relies heavily on its network infrastructure for daily operations. The company's network supports various departments, including Sales, Marketing, and Human Resources. The management is concerned about network performance, security, and potential issues that may disrupt business activities. Your task is to implement a robust network monitoring system to address these concerns and ensure the network's optimal functionality.

Required:

a) Describe the types of network monitoring tools you would consider implementing in this scenario. What specific functionalities should these tools possess to effectively monitor the network? **(6 Marks)**

b) Identify and explain at least three key performance indicators (KPIs) that you would monitor to assess the overall health and performance of the company's network. How do these KPIs contribute to proactive issue detection and resolution? **(5 Marks)**

c) Discuss the importance of integrating security monitoring into the network monitoring system. What specific security metrics or events would you track to ensure the network's resilience against potential threats? **(6 Marks)**

d) Explain the significance of historical data analysis in network monitoring. **(3 Marks)**

[Total: 20 Marks]

QUESTION 3

You are tasked with designing and configuring a network for a small startup company. The company has two departments: Sales and Development. The Sales department requires reliable access to a customer database, while the Development department needs a secure environment for collaborative coding projects. Additionally, the company wants a guest network for visitors but with limited access to company resources. Your goal is to create a network design that meets these requirements.

Required:

a) Describe the network topology you would recommend for the startup company, considering the needs of the Sales and Development departments. What factors influenced your choice of topology? **(5 Marks)**

b) Explain the process of subnetting and how you would allocate IP addresses to the Sales, Development, and Guest networks. What are the advantages of using subnetting in this scenario? **(5 Marks)**

c) Outline the security measures you would implement to protect the customer database in the Sales department and the collaborative coding projects in the Development department. **(5 Marks)**

d) Propose the use of VLANs to segregate the Sales, Development, and Guest networks. What benefits does VLAN implementation bring to network management and security? **(5 Marks)**

[Total: 20 Marks]

QUESTION 4.

Imagine you are the network administrator for a medium-sized company planning to upgrade its network infrastructure. The current network lacks efficiency, and there is a need for better connectivity, security, and management. Your task is to design and implement a network using various network devices to meet the company's requirements.

Required:

a) Explain the role and functionality of routers in a computer network. How can a router facilitate communication between different subnets within the company? **(5 Marks)**

b) Compare and contrast the functionalities of a switch and a hub. In what scenarios would you prefer to use a switch over a hub, and vice versa? **(5 Marks)**

c) Discuss the importance of a firewall in a network. How does it enhance the security posture of the company's network, and what types of traffic does it typically filter? **(5 Marks)**

d) Describe the role of a bridge in a network. How does it contribute to improving network performance? **(5 Marks)**

[Total: 20 Marks]

QUESTION 5.

You are working as an intern at a telecommunications company, and your supervisor has assigned you a project related to multiplexing and demultiplexing. The company is planning to upgrade its communication infrastructure to accommodate the increasing demand for data transmission. Your task is to design a multiplexing and demultiplexing system for a new communication network.

Required:

a) Briefly explain what multiplexing and demultiplexing are in the context of communication networks. **(5 Marks)**

b) Describe at least two types of multiplexing techniques commonly used in communication systems. Provide examples of situations where each type is most suitable. **(5 Marks)**

c) Discuss the advantages of using multiplexing in a communication network. How does it help in optimizing resource utilization and improving overall efficiency? **(5 Marks)**

d) Explain the demultiplexing process in detail. How does the system distinguish between different signals and route them to their respective destinations? **(5 Marks)**

[Total: 20 Marks]

END OF SEMESTER FINAL EXAMINATION